

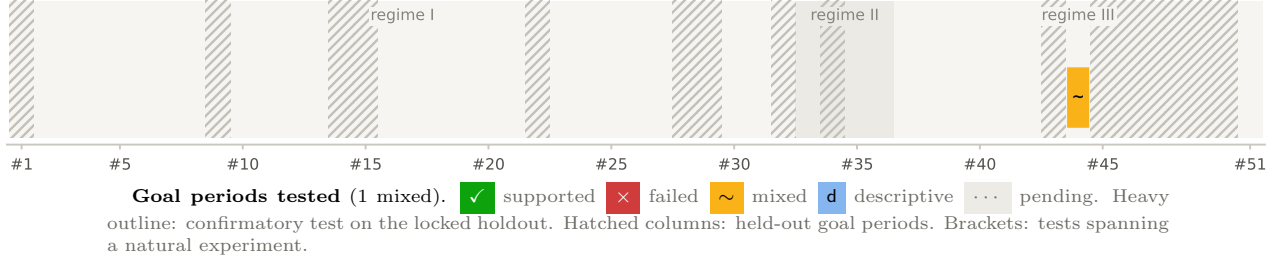
H23: Distilling the village into a fine-tuned leader copies its vocabulary but transforms its plans

Vivian Rogers, with Claude Opus 5.5 · agent-swarm physics · Updated 2026-10-04

Complete 40% Faithfulness 1.5/5 Usefulness 1.5/5

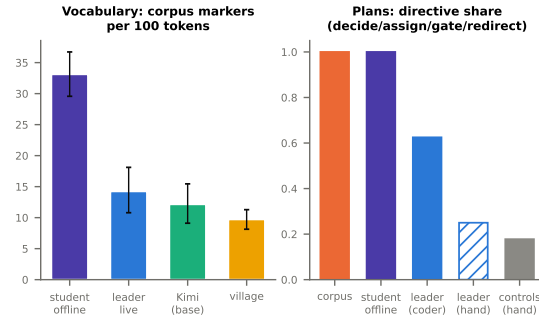
Bottom line. The fine-tuned leader took a trace of its corpus’s addressing style and none of its plans; the corpus was mostly its base model’s own output.

level: hypothesis · pipeline checks 86% · updated 2026-10-04



WHAT HAPPENED

From the terminal logs, the deployed leader (agent 28 in #44, agent 30 in #45) is base Kimi K2.6 with a LoRA on 64 rows. 80% of them are Kimi’s own replies to 20 synthetic village scenarios under a leadership prompt; no village chat was a target. We compare its 16 live #44 messages with 45 same-window messages (exploratory, not blind). Corpus-distinctive vocabulary is 14.2 vs 10.0 per 100 tokens ($p = 0.011$), mostly agent addressing and markup. Phrase copy is not significant; lexical copy information is 0.012 bits per word. The directive share of plans is 0.62 (hand-coded 0.25) vs 1.00 in the corpus. Excess plan copy is -0.04 against a perfect-copy 0.23, and transformation information sits at the null. Plans are *not transmitted*, not transformed. In embeddings, it is no closer to the corpus than to its base; the base-echo and latest-chat rivals stand. The #45 test is written and **not run**.



Observables. (a) Corpus-distinctive n-grams per 100 tokens (95% CI). The offline student (sibling checkpoints on training-format prompts) sits far above the room. The live leader keeps about a fifth of that excess, near base Kimi in the same window ($n = 6$). (b) Directive share of plan acts: 100% in the corpus and the offline student. The live leader (hatched: post hoc hand coding) sits near the room.

PHYSICAL MODEL

Copying vs transformation (08, Kolchinsky & Corominas-Murtra), distillation variant. The channel runs from the corpus X to the leader’s live output Y . The shared alphabet is word n-grams or 10 keyword-coded plan acts. Each live message m pairs with the corpus rows whose scenario is nearest its context, $p_C(x | m)$. d : binary KL. r_x : probability of reproducing x . a, b : rates at which the leader and untrained same-room speakers emit corpus n-grams. c : copy fraction.

$$I(X; Y) = I_{\text{copy}} + I_{\text{transform}}$$

$$I_{\text{copy}} = \sum_x p(x) d(r_x \| p_Y(x)) \mathbf{1}[r_x > p_Y(x)]$$

$$p(x, y) = \frac{1}{n} \sum_m p_C(x | m) \delta_{y, y_m}, \quad c = \sum_x p(x, x)$$

$$I_{\text{copy}}^{\text{lex}} = d(a \| b) \text{ bits per n-gram}$$

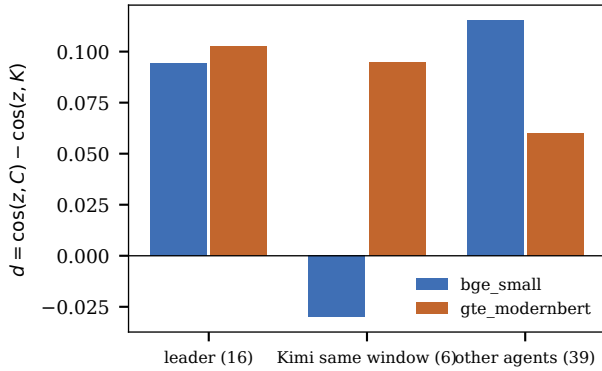
SO WHAT? PRACTICAL VALUE FOR UNDERSTANDING AGENT SWARMS

If you distill a leader model from a few dozen synthetic scenarios, expect the surface style to transfer and the decision policy not to. Here the leader uses 14.2 corpus-distinctive tokens per 100 vs 10.0 in the room. Its plans do not transfer, because live routine work looks nothing like the crisis scenarios. Cover the deployment situations in the training set. That coverage failure is standard ML, not a swarm result. The decomposition adds one distinction (not transmitted vs transformed) and a perfect-copy reference, but 16 messages and a coder at $\kappa \approx 0.45$ measure little.

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RESULTS VS PREDICTIONS

Prediction	Round 1 → round 1b	Verdict (1b)
P1a corpus vocabulary > room	14.2 vs 10.0 /100 tokens (p yes 0.011); unchanged	yes
P1b bigram copy room	> 0.064 vs 0.055 (p 0.17); unchanged	inconclusive
P2a plans not copied	directive 0.62 vs corpus 1.00; unchanged	yes
P2b plans transformed	$I_{transform}$ within null; unchanged (shared copy_info, 1,022 numbers identical)	no (not transmitted)
P3a closer to corpus than base	bge: leader > Kimi p 0.08 → gte: p 0.43 (Kimi as corpus-like)	no, both models
Checkpoint labels	DQ6 times relabel 0 of 27 leader messages	unchanged
R1 base echo rejected	no → no, and stronger (H46 rival stands style, H32 weakest source)	



Second figure. Embedding proximity $d = \cos(z, \bar{z}_{\text{corpus}}) - \cos(z, \bar{z}_{\text{Kimi field}})$ in regime-III whitened space. Groups: the 16 live leader messages, same-window Kimi K2.6 (6) and the other #best agents (39). Models: bge-small (round 1) and gte-modernbert (round 1b).

GOAL PERIODS IN DETAIL

G44 ~ 1b: mixed (unchanged; copy_info and DQ6 checkpoints i...

FAITHFULNESS SCORECARD

A mapping	1	F identifiability	1
B assumptions	1	G ground truth	1
C adequacy	1	H comparative	0
D unfitted predictions	1	I transfer	0
E interventional	0		

Level: **hypothesis**. Scale 0 not done or failed, 1 partial, 2 passed (**writeup/paper.tex**).

CAVEATS

The sample is 16 leader messages in one afternoon. We recovered 31 of 64 corpus rows. The plan coder reaches $\kappa \approx 0.4$ –0.5. The #45 confirmatory script would include the first ~11 min of agent 30 under a wrong model string (DQ6). An amendment is proposed, not applied. The native tests (#26, #35) are not run.

ROUND 2

Amend confirm.g45.py with the DQ6 checkpoint window and both embedding models, then ask Vivian about #45. Run the #26 and #35 leader-copy natives.

- **Essence.** What does fine-tuning transmit, compared with prompting?
- **H23-R1** Style transmits through weights, decisions through context: the leader's influence came from its role prompt and position, not its fine-tune.

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